OMEX Experiment Report — vcornea_sls_sweep_20261020_182006

3 combinations | 2, 2, 2 replicate(s) each | SimTime 30 days | injury at day 3.0

Experiment Archive

This report describes the experiment-level archive **vcornea_sls_sweep_scan.omex** (242,194 bytes) — the SED-ML/COMBINE description of the whole scan. The per-combination reports describe each combination's per-replicate archives; this one sits above them and compares the combos.

Parameter scan	SLS_Concentration ∈ [250.0, 750.0, 2000.0]
Scan points	3
Replicates / point	2
Total simulations	6

Experiment OMEX Contents

File	Size (B)
combo_data/vcornea_sls_sweep_combo001_cell_count.csv	19,391
combo_data/vcornea_sls_sweep_combo001_surfactant.csv	170,780
combo_data/vcornea_sls_sweep_combo001_thickness.csv	18,078
combo_data/vcornea_sls_sweep_combo002_cell_count.csv	19,391
combo_data/vcornea_sls_sweep_combo002_surfactant.csv	170,711
combo_data/vcornea_sls_sweep_combo002_thickness.csv	18,069
combo_data/vcornea_sls_sweep_combo003_cell_count.csv	19,358
combo_data/vcornea_sls_sweep_combo003_surfactant.csv	170,573
combo_data/vcornea_sls_sweep_combo003_thickness.csv	18,087
manifest.xml	1,662
metadata.rdf	1,584
scan.sedml	29,555
scan_metadata.json	1,355

Reading the SED-ML (comparison)

scan.sedml encodes the scan as a repeatedTask AND compares the combinations: 9 dataDescriptions (one per combo × metric, bundled under combo_data/) feed 6 plot2D outputs — each overlays the combinations for one signal — and a 18-series report.

plot2D (signal across combinations)	curves (= combinations)
Superficial (cell_count) across combinations	3
Wing (cell_count) across combinations	3
Basal (cell_count) across combinations	3
Stem (cell_count) across combinations	3
Total Amount (surfactant) across combinations	3
Height (thickness) across combinations	3

Provenance (OMEX Metadata)

Field	Value
Creator (experimenter)	Joel Vanin
Created	2026-10-21T01:01:48
Archived	2026-10-21T01:01:48
License	https://spdx.org/licenses/CC-BY-4.0.html
Citation	Vanin et al. (2025). V-Cornea: a computational model of the corneal epithelium.
Keywords	CompuCell3D, cornea, parameter scan, agent-based model, wound healing

From metadata.rdf in the scan archive: attribution / provenance, not authentication.

Validation of the Experiment Archive

Layer	Result
Built-in SED-ML structure	PASS
Built-in OMEX manifest	PASS
biosimulators-utils (COMBINE)	PASS (0 errors, 1 warning group)

Combinations

Combination	Label	Replicates	Unsuccessful
vcornea_sls_sweep_combo001	SLS_Concentration=250.0	2	0
vcornea_sls_sweep_combo002	SLS_Concentration=750.0	2	0
vcornea_sls_sweep_combo003	SLS_Concentration=2000.0	2	0

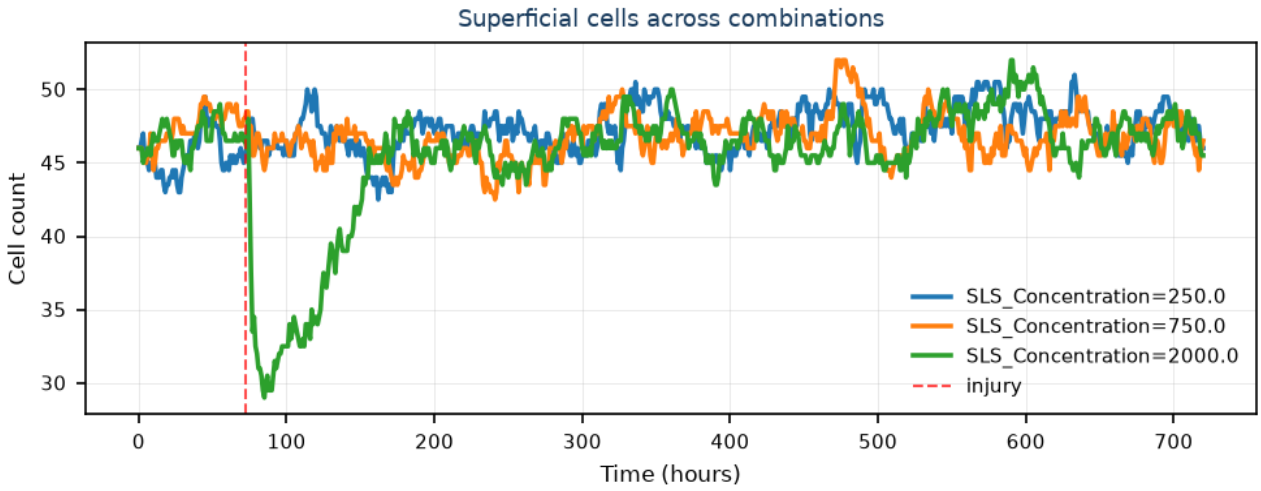
Summary Metrics (from replicate-mean series)

Combination	Wing baseline	Wing min	Wing final	Recovery %	Peak surfactant	Central thickness min (μm)	Central thickness final (μm)
SLS_Concentration=250.0	120.4	119.0	131.0	108.8	250.0	38.7	42.6
SLS_Concentration=750.0	116.5	112.0	128.5	110.3	750.0	38.3	42.5
SLS_Concentration=2000.0	116.8	87.5	125.5	107.4	2000.0	11.3	40.8

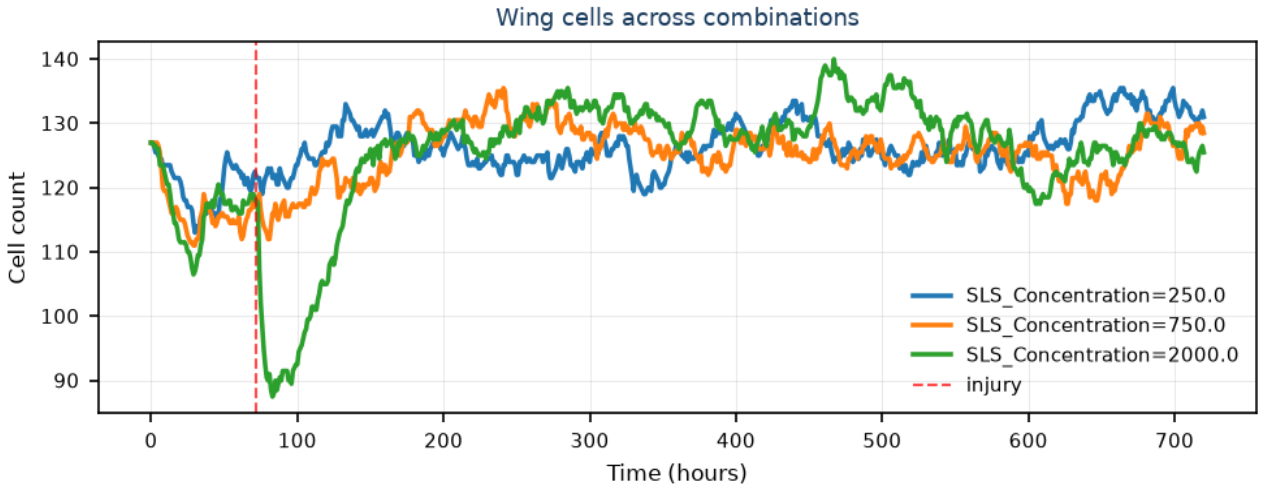
Wing baseline = pre-injury mean; Wing min = minimum after injury (damage depth); Recovery % = final / baseline. Central thickness = mean over bins 4–9, minimum after injury and at the end of the run. Metrics are on the across-replicate mean.

Metric Comparisons (one line per combination)

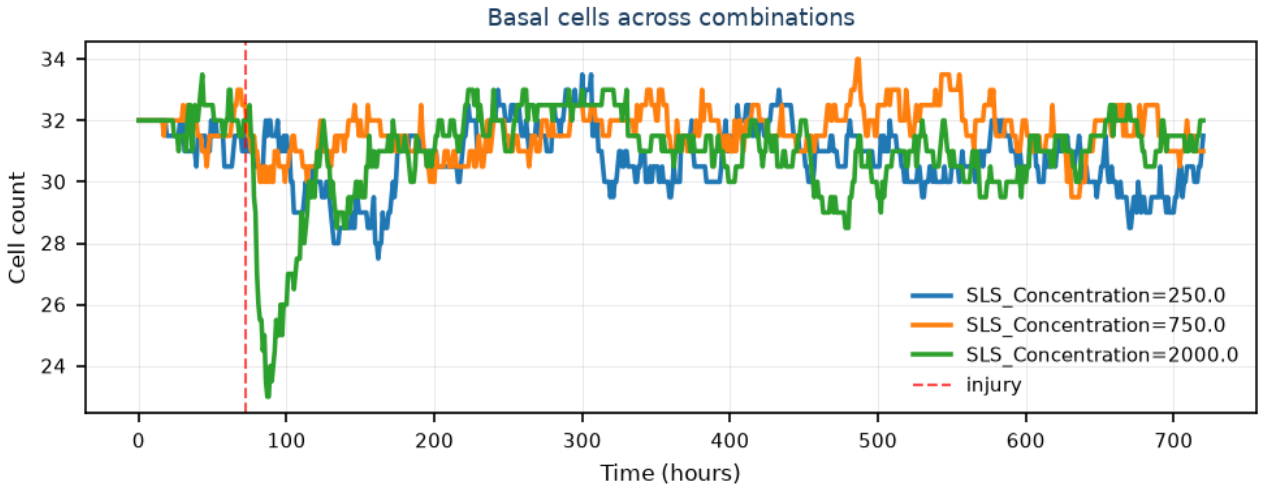
Each line is a combination's across-replicate mean, read from the per-run OMEX archives. Time in hours (thickness: days); the dashed red line marks injury.



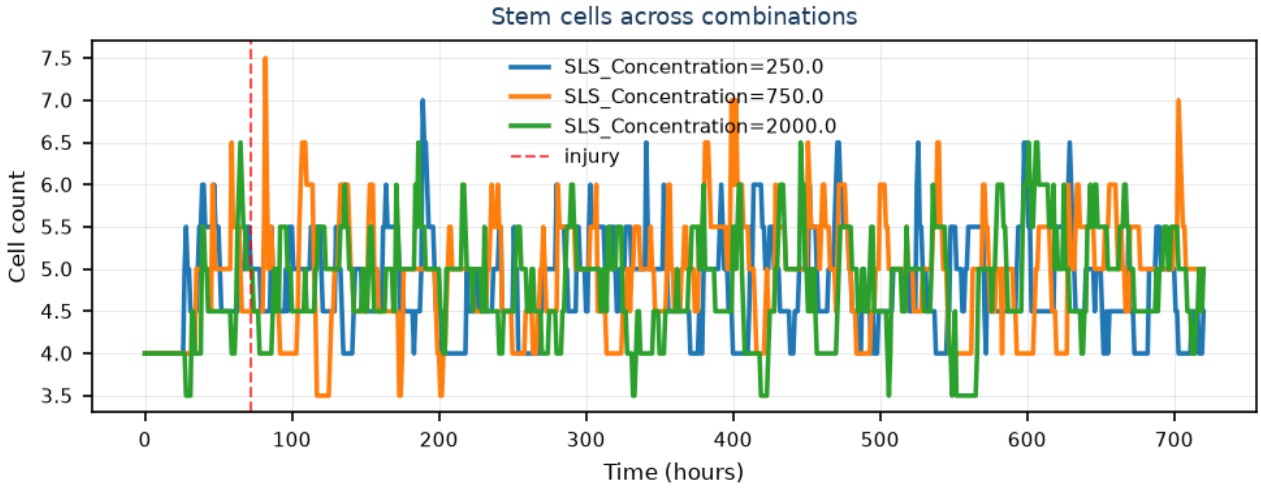
Superficial: replicate-mean count per combination vs Time



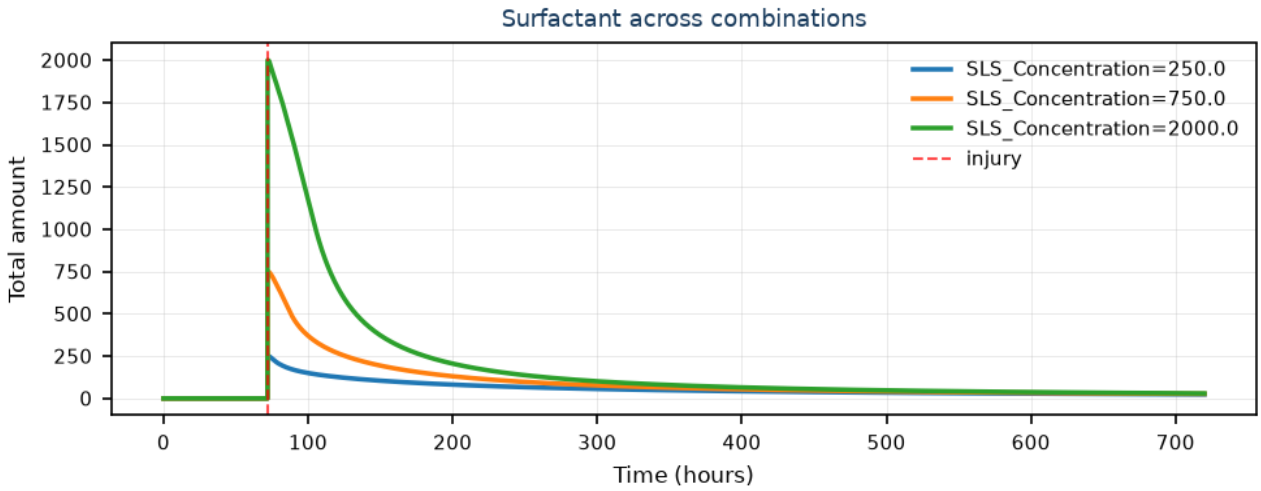
Wing: replicate-mean count per combination vs Time



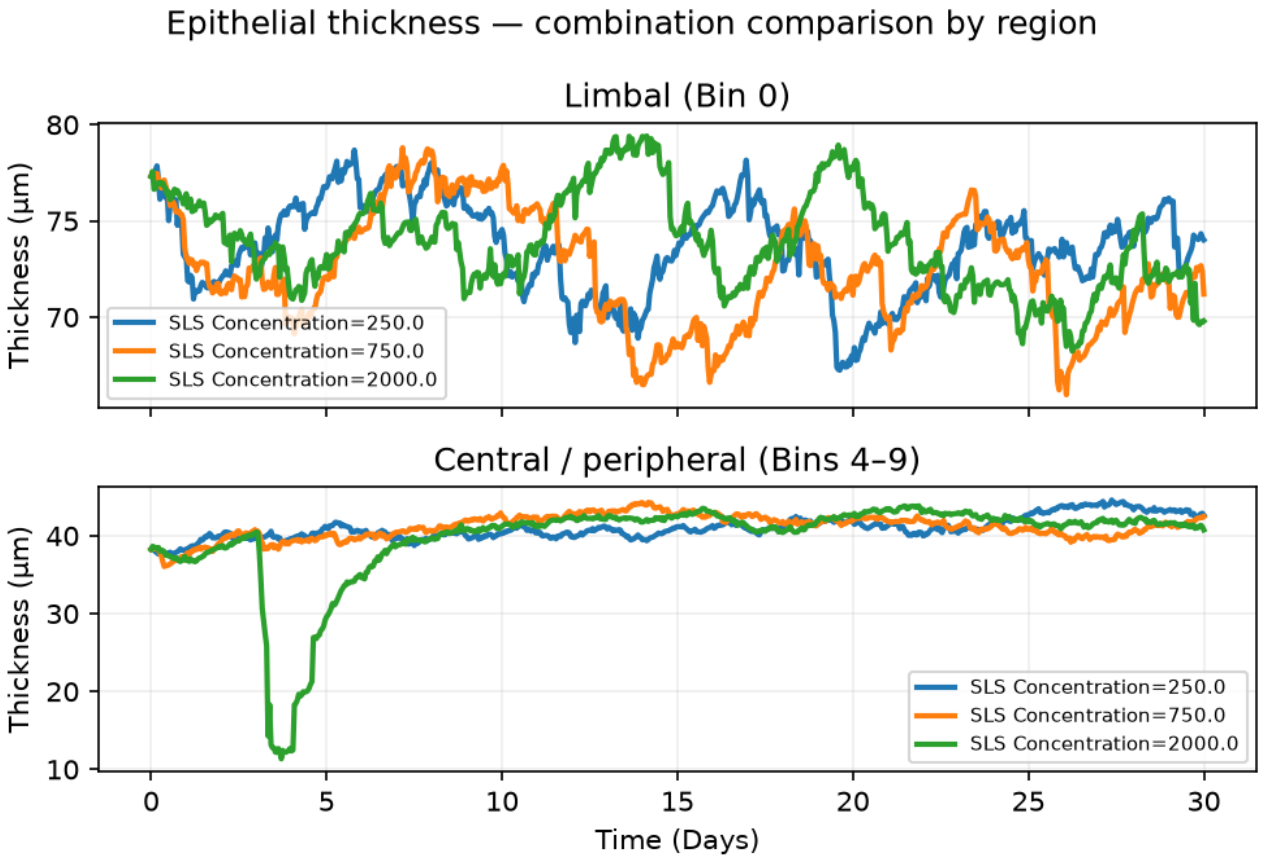
Basal: replicate-mean count per combination vs Time



Stem: replicate-mean count per combination vs Time



Surfactant Total Amount: replicate-mean per combination vs Time



Epithelial thickness in µm over time in the limbus (bin 0) and the central/peripheral cornea (bins 4–9), one line per combination (replicate mean), drawn by the V-Cornea GUI's own analysis (Analyze Results → Thickness, comparison view). Thickness = (top position – membrane baseline) × 2 µm/voxel. Time in days.